Programming Languages Lab
CS 331

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Task - shoot yourself at the foot ☺

(Compiled from various versions available on the web)

C   JAVA   PYTHON PROLOG   C++   HASKEL

You shoot yourself in the foot and then nobody else can figure out what you did.

You accidentally create a dozen instances of yourself and shoot them all in the foot. Providing emergency medical assistance is impossible since you can't tell which are bitwise copies and which are just pointing at others and saying, "That's me, over there."

You write a program to shoot yourself in the foot and put it on the Internet. People all over the world shoot themselves in the foot everyone leaves your website hobbling and cursing. You amputate your foot at the ankle with a fourteen-pound hacksaw but you can do it on any platform.

You insert the bullet into the gun but because there were actually TAB characters in the barrel instead of spaces, the gun jams with an IndentError. You fix the error and fire the gun, but the bullet travels so slowly towards your foot that you get tired and leave.

You shoot yourself in the foot very elegantly, and wonder why the whole world isn't shooting itself this way.

You tell your program you wish to be shot in the foot. The program figures out how to do it, but the syntax doesn't allow it to explain.

Introduction

- What this course is all about?

  - introduces multiple programming paradigms focuses on how different languages express computation
  - hands-on experience with:
    - Imperative and declarative styles
    - Concurrent programming using Java threads and libraries
    - Logic programming using Prolog
    - Functional programming using Haskell

Introduction

- What will you be able to do by the end of this course?
  - Understand and compare **different programming paradigms** and their use cases
  - Write and execute **concurrent programs** using Java threads and synchronization mechanisms
  - Model and solve problems using **logic programming** in Prolog
  - Develop programs using **functional programming concepts** such as recursion, immutability, and higher-order functions
  - Choose an appropriate programming paradigm for a given problem

Evaluation

- Logic Programming : 3 Assignments
- Functional Programming : 2 Assignments
- Concurrent Programming : 2 Assignments

- Assignments **70%** + Assessment **30%**

Roadmap

- January: Logic Programming
  - 3 Tutorials, 1 Assignments
- February: Logic Programming + Concurrent Programming
  - 4 Tutorials, 2 Assignments
- March: Concurrent Programming + Functional Programming
  - 3 Tutorials , 2 Assignments
- April : Assessment + Functional Programming
  - 1 Tutorial , 2 Assignments

Schedule

Date	Tutorial	Assignment
7 Jan	Introduction to course & Tutorial 1 – Logic Programming (LP)	
21 Jan	Tutorial 2 – Logic Programming	Assignment 1 Due
28 Jan	Tutorial 3 - Logic Programming	
4 Feb	Tutorial 4- Logic Programming	Assignment 2 Due
11 Feb	Tutorial 5- Concurrent Programming	
18 Feb	Tutorial 6 – Concurrent Programming	Assignment 3 Due
25 Feb	Tutorial 7- Concurrent Programming	
11 Mar	Tutorial 8 – Functional Programming	Assignment 4 Due
18 Mar	Tutorial 9 – Functional Programming	
25 Mar	Tutorial 10 – Functional Programming	Assignment 5 Due
1 April	Assessment	
8 April	Assessment	
22 April	Tutorial 11 (Optional)	Assignment 6 Due
29 April	--	Assignment 7 Due

Books

- Clocksin, W.F. and Mellish, C.S., Programming in Prolog
  - A good basic introductory book.
- Bratko, I., Prolog Programming for Artificial Intelligence
  - An introductory book that leans heavily towards AI applications.
- Sterling, L. and Shapiro, E., The Art of Prolog
  - Possibly the best general Prolog book around, but definitely not for beginners.

Other Details

- Tutorial in Seminar Room (Old) – 11:00-12:00 (Wednesday) – Mandatory
- Note – allotted slot for Tutorial is D1 Wednesday 17:00 – 18:00 hrs.
- Lab hours : 9:00 – 12:00 (Wednesday) – Not mandatory to be in the Lab.
- TAs for the course:
  - Nirban Das (d.nirban@iitg.ac.in)
  - Rahul Pandey (rahul.pandey@iitg.ac.in)
  - Archana K R (r.archana@iitg.ac.in)